AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 6 · WORKBOOK

Debugging Strategies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected function-testing record.

Derive expectations, expose a constant answer, repair addition and test the new stock formula before judging its limits.

Keep one habit: Keep input, expected value, actual value, comparison, version and regression evidence together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Loops and Stopping

Branching Logic

MAIN LESSON · SECTION 12

W1 · Derive the six-capacity expectations

> Use the supplied function and rule reference.

Your evidence record

- For W1–W3 use the same remaining(capacity,used)=capacity-used rule, now with capacity 6. Calculate expectations before running. Keep the constant-4 and addition bodies as separately labelled faulty versions; they do not change the expected answers.

- Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

```
def remaining(capacity, used):
    return capacity - used

cases = [(8, 3, 5), (8, 8, 0), (8, 0, 8)]
for capacity, used, expected in cases:
    actual = remaining(capacity, used)
    print(actual == expected)
```

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Test the constant-four defect

> State which particular expected values it accidentally matches.

Your evidence record

- A faulty function always returns 4. Which of your three cases pass and fail?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair addition to subtraction

> Preserve the old run and choose a prior test to rerun.

Your evidence record

- A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a **regression** case.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Test a new function independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Test a new function independently

MAIN LESSON · SECTION 16

W5 · Limit the independent stock-test conclusion

> After W4, distinguish valid test passes from untested negative counts.

Your evidence record

- All three stock tests pass. Does that prove correct handling of negative counts?

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

When a ranking surprises you, inspect feature values, weights, arithmetic, and sorting separately.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Find a revealing case for Testing and **debug**ging functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Testing and **debug**ging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Award two points per matching trait. A bug awards two for every trait.

Test one match and one nonmatch.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 6 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional shelf function should count books in complete bundles plus loose books.

Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose.

Faulty copied body returns (bundle_count+loose)*6. Predict tests before running.

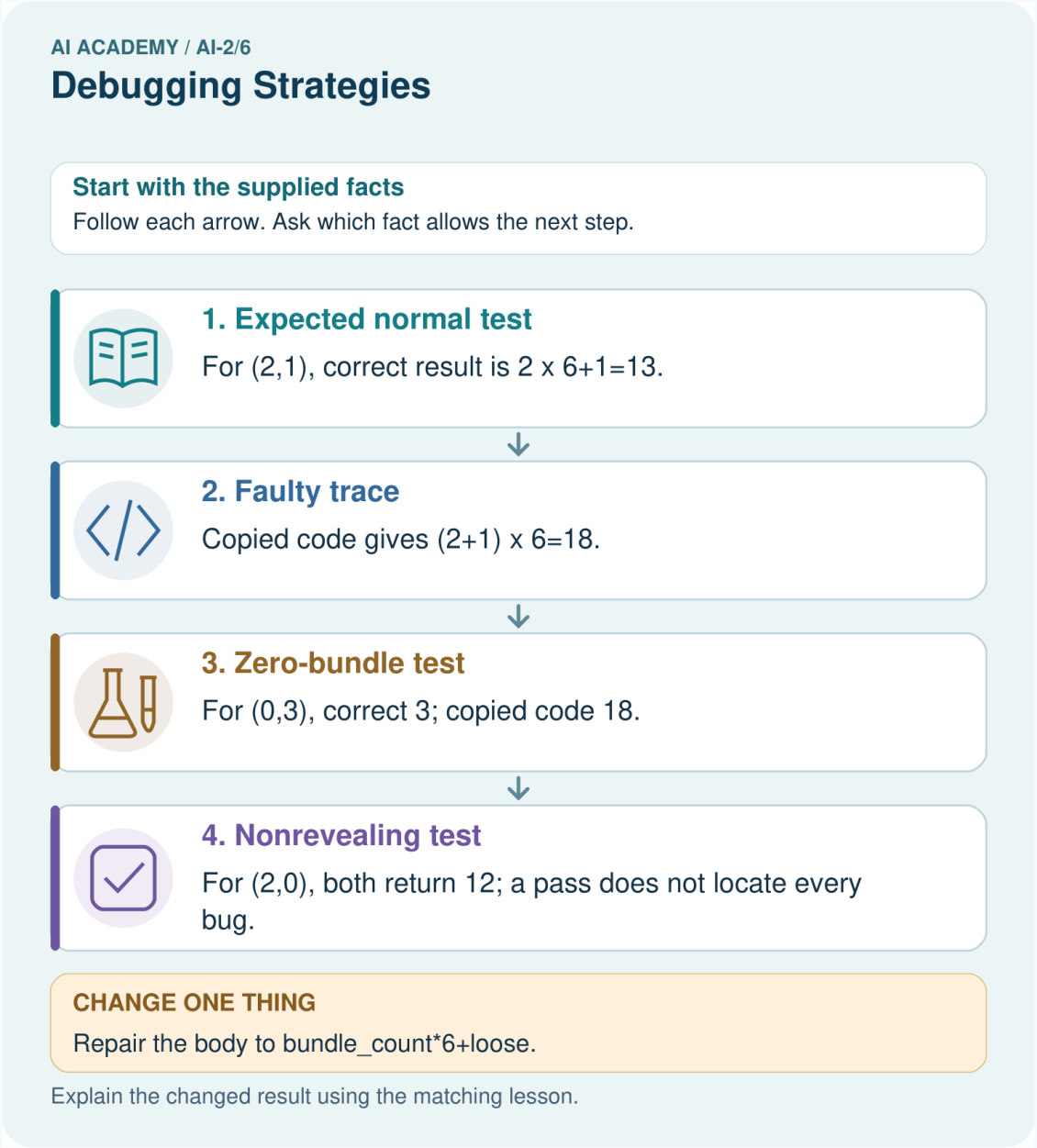
AI ACADEMY / AI-2/6

Expected outputs versus a specific bug

	Bundles	Loose	Correct	Faulty
Test 1	2	1	13	18
Test 2	0	3	3	18
Test 3	2	0	12	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which test most directly isolates the handling of loose books, and what should it produce?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Test the recommender’s helper functions with normal, zero and **boundary input**s before **debug**ging the whole application.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Test the recommender’s helper functions with normal, zero and **boundary input**s before **debug**ging the whole application.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional shelf function should count books in complete bundles plus loose books.
- Which labelled step matches this detail? For (2,0), both return 12; a pass does not locate every bug.
- 1. Zero-bundle test
- 2. Expected normal test
- 3. Faulty trace
- 4. Nonrevealing test
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Nonrevealing test For (2,0), both return 12; a pass does not locate every bug. **Test case**s need not all expose the same error.

Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

Changed-case question

- Changed-case practice: Repair the body to bundle_count*6+loose. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Expected results for (2,1),(0,3),(2,0) are 13,3,12.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Expected results for (2,1),(0,3), (2,0) are 13,3,12. Only bundles are groups of six; loose books are counted individually.

Repair and save your work

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short..

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

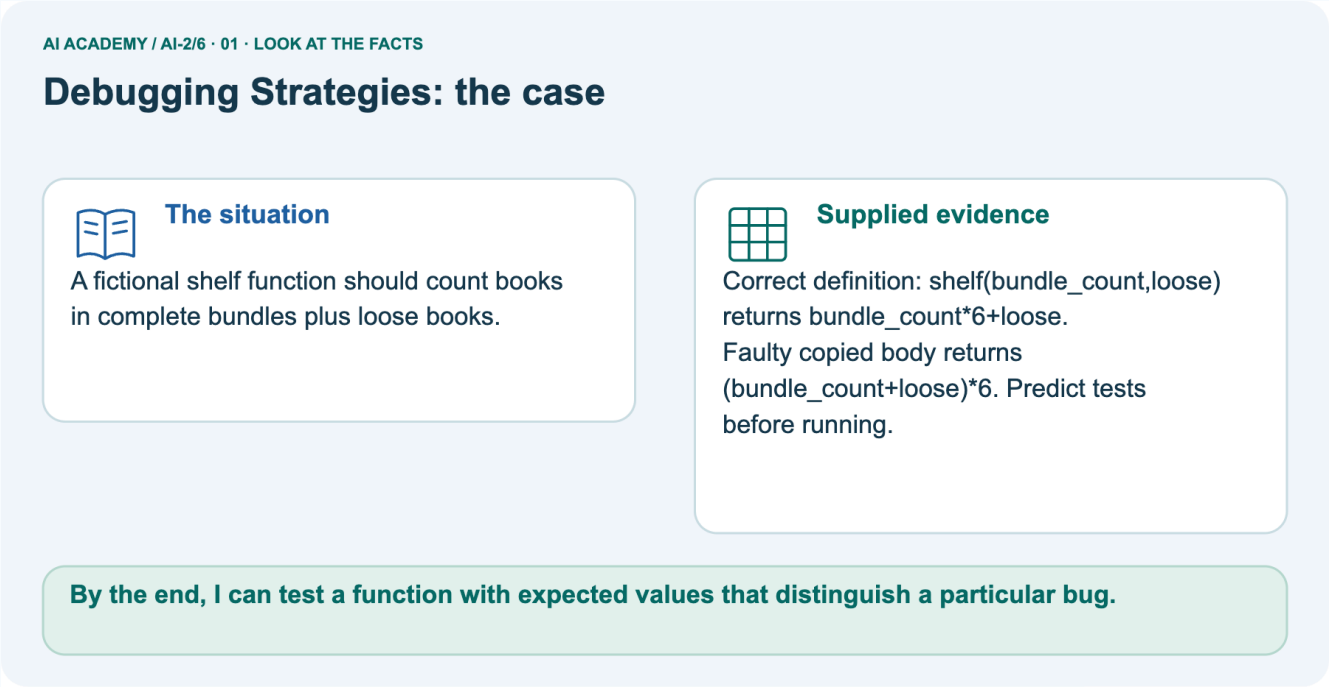
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Debugging Strategies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

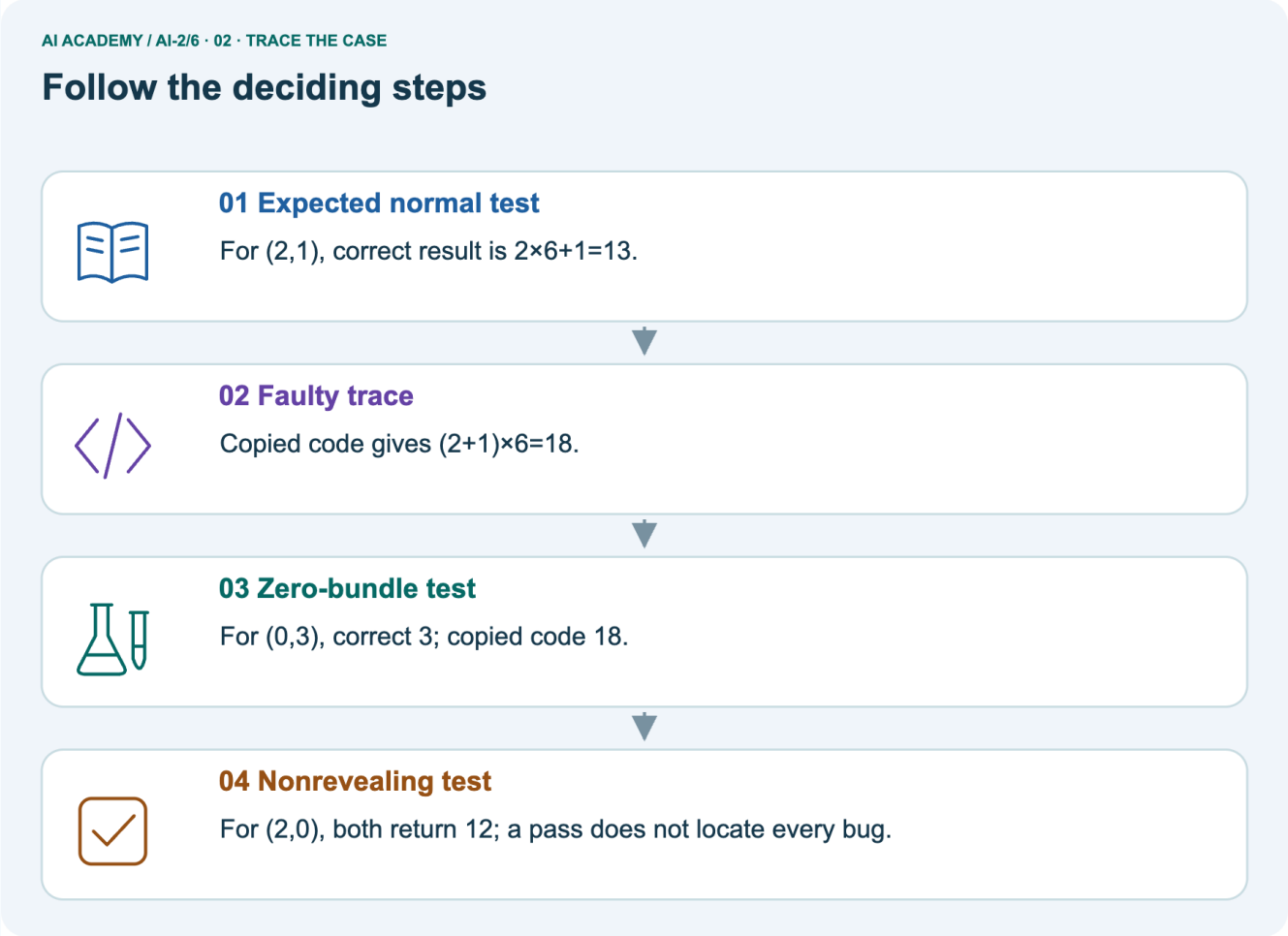
A fictional shelf function should count books in complete bundles plus loose books. Supplied facts: Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose. Faulty copied body returns (bundle_count+loose)*6.

Predict tests before running.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



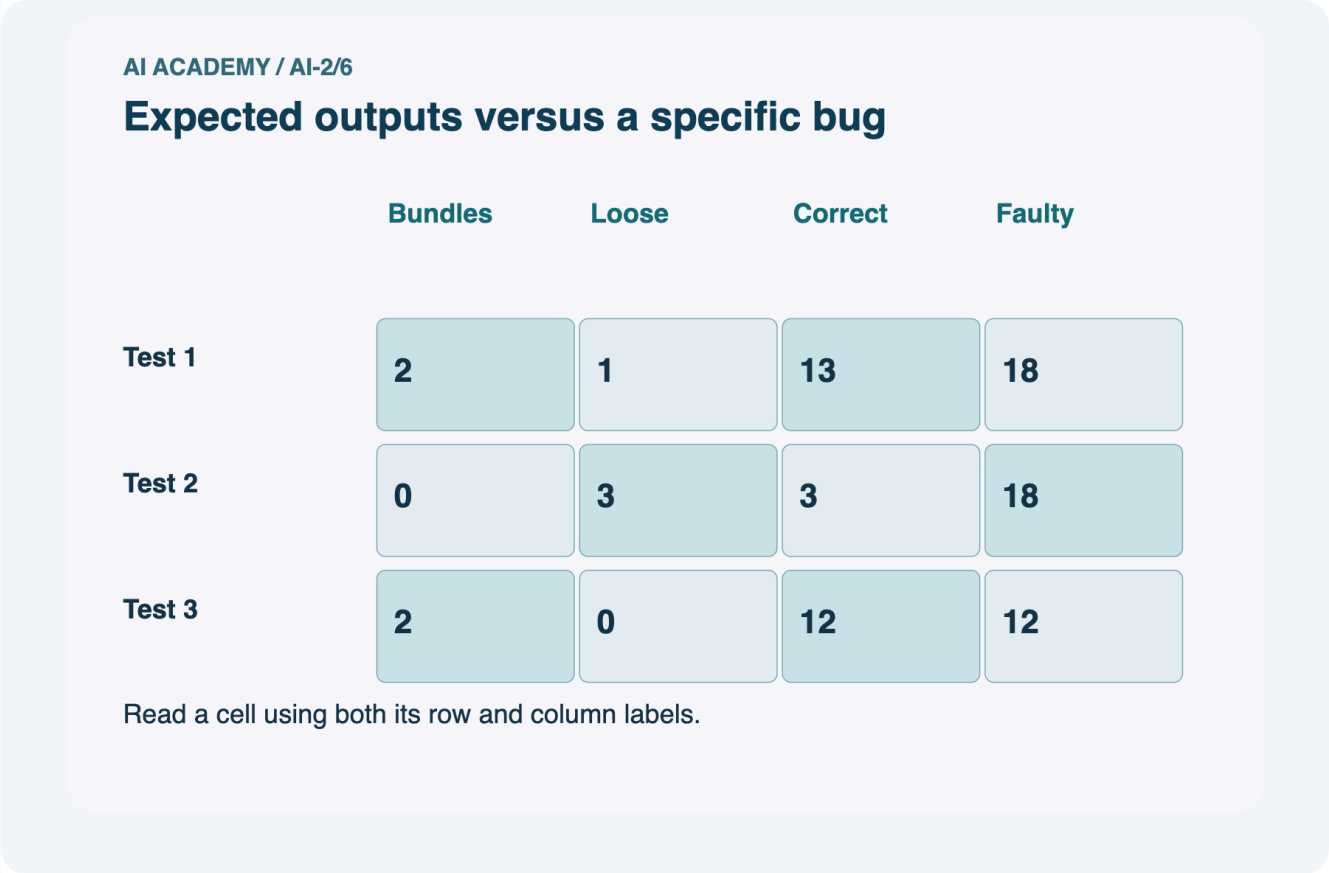
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected normal test: For (2,1), correct result is 2×6+1=13. → Faulty trace: Copied code gives (2+1)×6=18. → Zero-bundle test: For (0,3), correct 3; copied code 18. → Nonrevealing test: For (2,0), both return 12; a pass does not locate every bug.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Expected outputs versus a specific bug



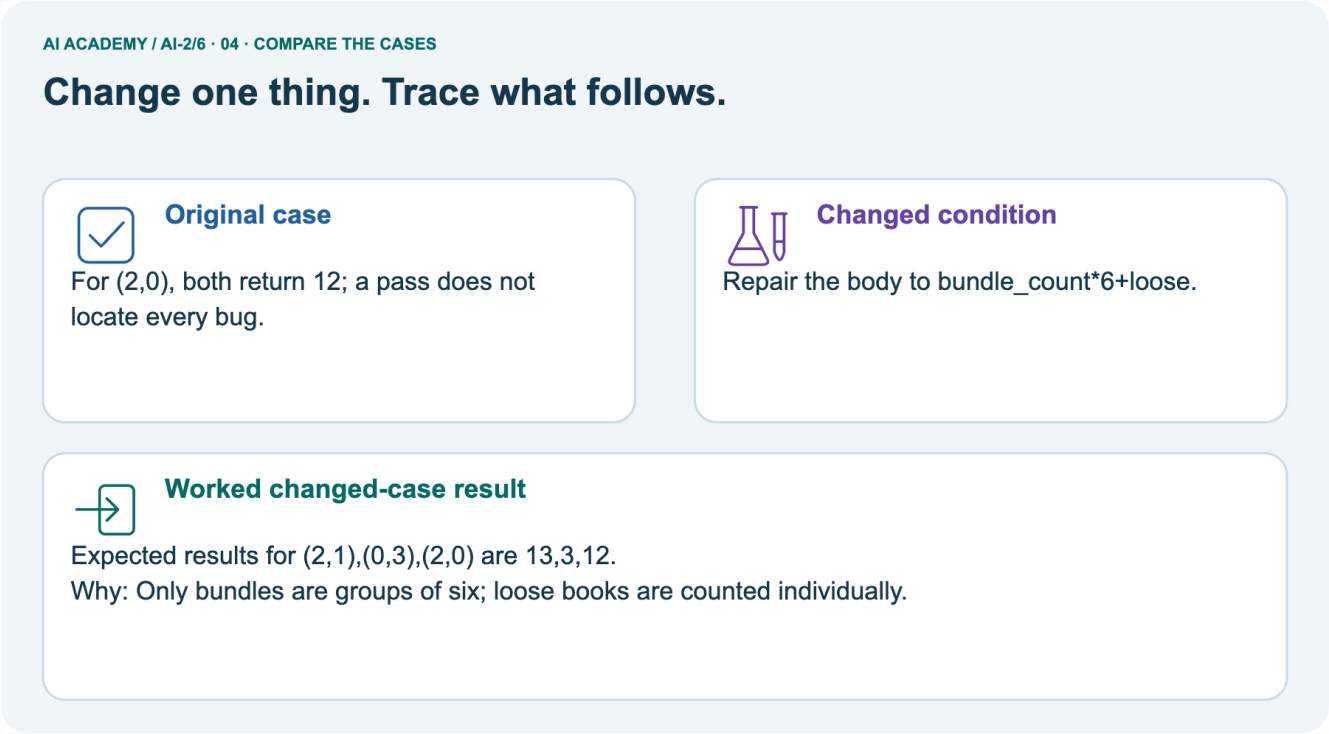
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected **output**s versus a specific bug. Test 1: Bundles = 2, Loose = 1, Correct = 13, Faulty = 18. Test 2: Bundles = 0, Loose = 3, Correct = 3, Faulty = 18. Test 3: Bundles = 2, Loose = 0, Correct = 12, Faulty = 12.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

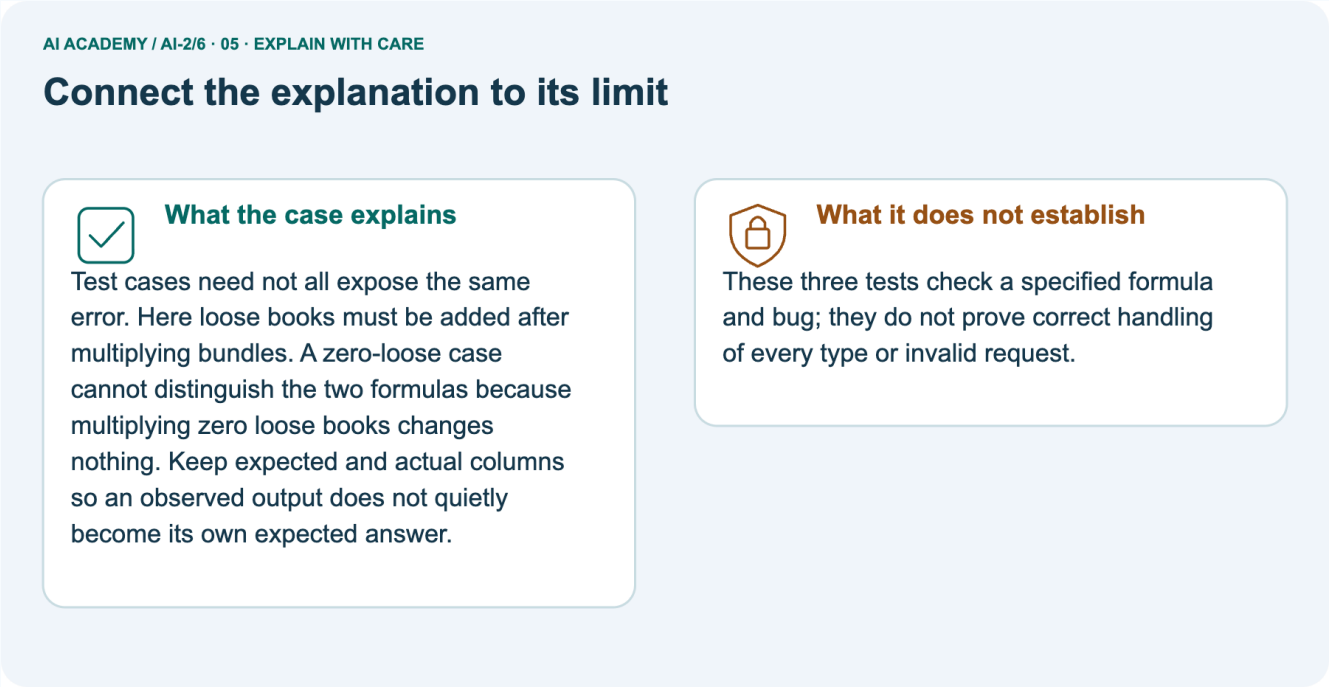
Original case: For (2,0), both return 12; a pass does not locate every bug. Changed condition: Repair the body to bundle_count*6+loose. Expected result: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

Reason: Only bundles are groups of six; loose books are counted individually.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

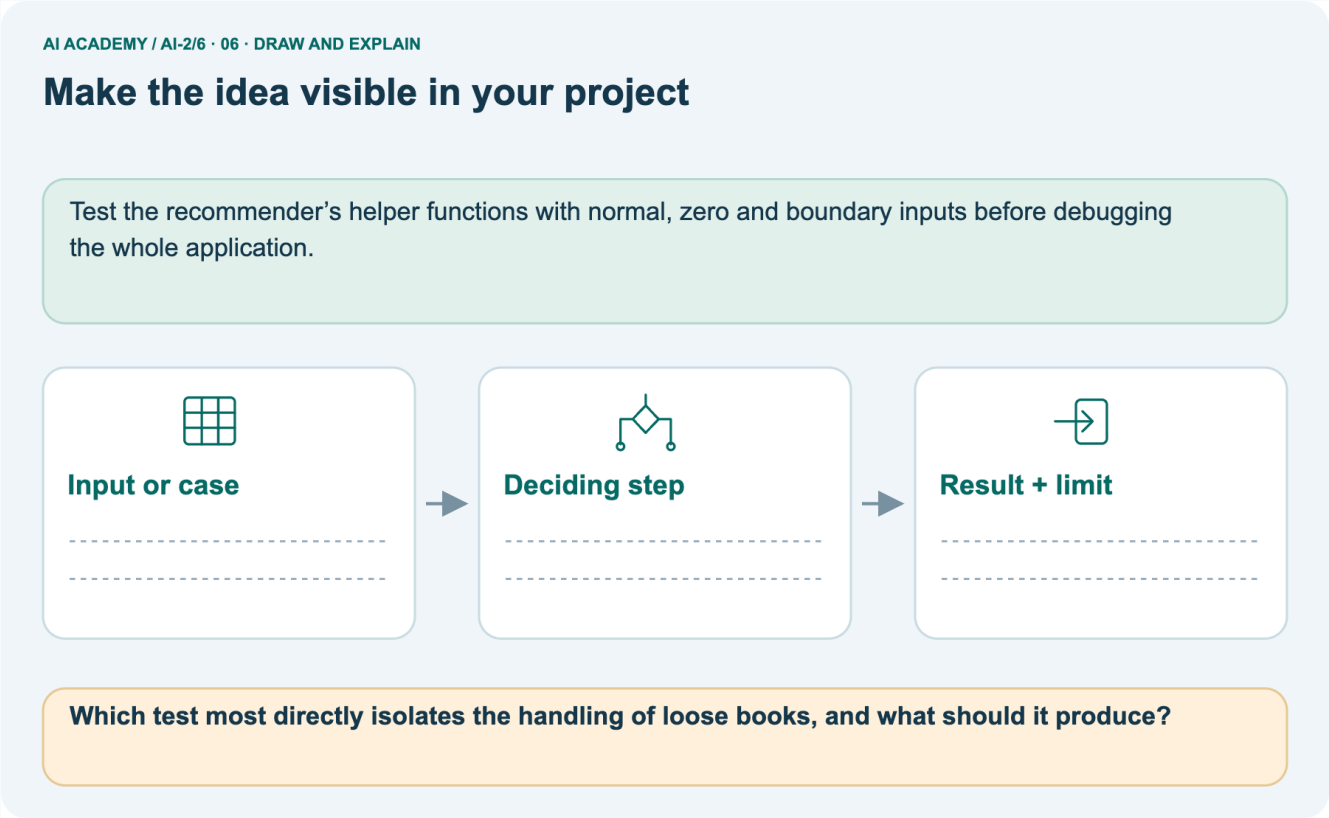
Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

Limit: These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test the recommender’s helper functions with normal, zero and **boundary input**s before **debug**ging the whole application. Draw your **input**, deciding step, result and limit.

Supported question: Which test most directly isolates the handling of loose books, and what should it produce?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Debugging Strategies: the case

02. Follow the deciding steps

03. Expected outputs versus a specific bug

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.